

Animex
Multimodal Interactive Technologies and Applications Project

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Contents

1	Project Resources and Installation Guide	3
1.1	Project Repositories	3
1.2	Usage Guide	3
1.2.1	Android	3
1.2.2	iOS	3
1.2.3	Additional Notes	3
2	Conceptual Description	4
2.1	System Architecture	4
2.2	Main Functional Modules	4
2.3	Circular Interaction Flow	4
3	Main Tasks	5
3.1	User Scenario	5
3.2	Scenario Analysis	5
3.3	Simple Task: Visual Identification	5
3.3.1	Hierarchical Task Analysis	5
3.3.2	Grounding and Error Recovery	6
3.4	Core Task: Questions and Quizzes	6
3.4.1	Hierarchical Task Analysis	6
3.4.2	Grounding and Error Recovery	6
3.5	Advanced Task: Natural-Habitat Immersive Experience	6
3.5.1	Hierarchical Task Analysis	7
3.5.2	Grounding and Error Recovery	7
3.6	Relationship Between the Tasks	7
4	Project Architecture	8
4.1	Front-end Application	8
4.1.1	Overview	8
4.1.2	Service Layer	9
4.1.3	Module Layer	9
4.1.4	Presentation Layer	9
5	Response to Heuristic Evaluation	10

1 Project Resources and Installation Guide

1.1 Project Repositories

Here you can find the links to the Animex Git repositories.

- Front-end application
- Back-end server

1.2 Usage Guide

To test the Animex application you must follow these steps:

1.2.1 Android

1. Go to the front-end repository.
2. Download the latest Animex.apk version from the releases section.
3. Install the APK on your Android device.
4. Run the application.

Fallback: Clone the repository, follow the repo's README to run the program through Android Studio.

1.2.2 iOS

1. Go to the front-end repository.
2. Clone the repository.
3. Follow the steps in the repo's README to run the program through XCode.

1.2.3 Additional Notes

Currently, the Animex system supports recognition for the following animal species:

- Dog
- Cat
- Lion
- Zebra
- Hippopotamus
- Tiger
- Frog

In the absence of a live specimen, the recognition also works on images.

2 Conceptual Description

Animex is an educational mobile application designed to bridge the gap between outdoor exploration and interactive learning.

Acting as a portable, AI-powered digital zoologist, the application aims to foster environmental awareness by transforming passive wildlife observation into an active and engaging experience.

Through the integration of computer vision, spatial computing, and natural language processing, Animex allows users to identify animal species, visualize them in their physical environment, and deepen their understanding through conversational interactions and gamification.

2.1 System Architecture

The application is built upon a modular and highly decoupled architecture. This design ensures a fluid and frictionless user experience, characterized by seamless navigation and dynamic interface adaptations.

A centralized state management system is employed to strictly separate the underlying business logic from the graphical user interface.

This conceptual separation allows the application to efficiently process real time multimodal data such as live camera feeds, voice inputs, and active learning sessions without compromising performance or visual continuity.

2.2 Main Functional Modules

The Animex ecosystem is conceptually divided into three interconnected macro-modules:

1. **Scanning and Recognition Engine:** This module leverages real time computer vision to identify wildlife. It features an immersive, distraction-free interface that dynamically adapts its layout to prioritize the camera feed during active scanning, ensuring an optimal framing and detection experience for the user.
2. **Conversational Assistant and Gamification:** Acting as the interactive core, this multimodal chatbot enables bidirectional communication via text or voice. It provides contextual educational insights based on the identified species.
To promote active learning, the module seamlessly integrates an adaptive quiz system. User can ask the system to generate a quiz, offering varying levels of difficulty (e.g., multiple-choice or true/false) and accepting answers through intuitive touch interactions, typing, or vocal commands.
3. **Virtual Reality (VR) Vision:** This module introduces spatial computing to the application, allowing users to project 3D models of the identified animals and their natural habitats directly into their surroundings. By providing a tangible sense of scale and anatomical detail, the VR module acts as a visual bridge between theoretical knowledge and the physical world, significantly enhancing user immersion.

2.3 Circular Interaction Flow

The conceptual strength of Animex lies in its continuous and circular usage flow. The visual detection of an animal instantly updates the session’s contextual knowledge base. This data seamlessly feeds into the AR module for spatial visualization and equips the virtual assistant to offer targeted educational content and quizzes.

This architecture creates a cohesive and engaging learning loop: *Detection* → *Visualization* → *Learning* → *Verification*.

3 Main Tasks

3.1 User Scenario

Giorgino is visiting a zoo with his family. As he walks through the zoo, he sees many animals that he has never encountered before and would like to learn more about their behavior, diet, routines, and natural habitats. Since Giorgino is a child, the interaction must be entertaining, immediate, and easy to understand. Moreover, because a zoo can be crowded and noisy, the application must present information clearly and concisely and must provide alternatives when speech interaction is unreliable.

3.2 Scenario Analysis

User goal: Identify unfamiliar animals and develop an understanding of their main traits, habits, routines, and natural habitats.

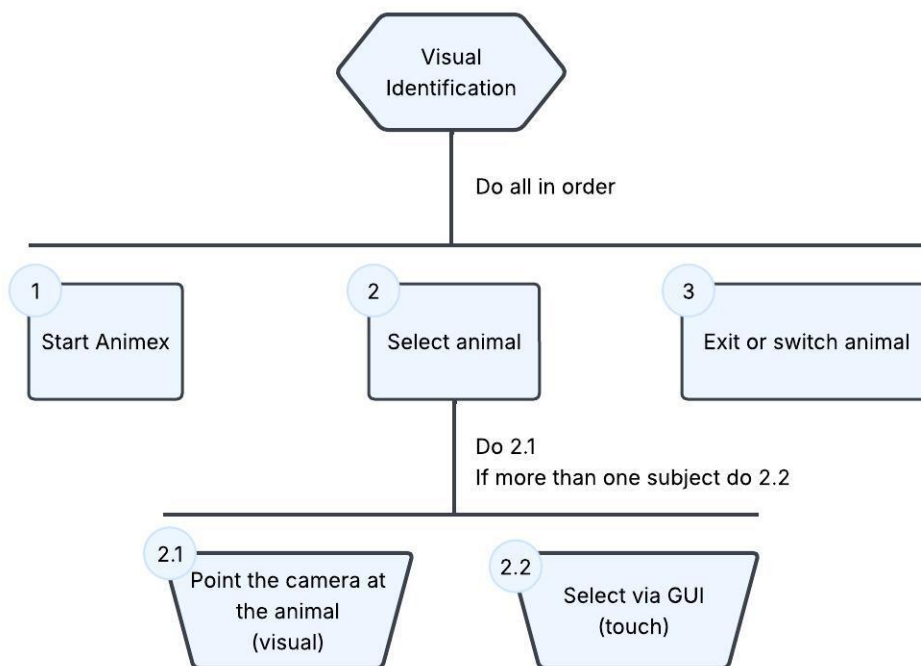
Environmental constraints: The interaction takes place outdoors while the user may be standing or walking and holding the device in one hand. Lighting conditions can vary, background noise may interfere with speech recognition, animals may move or be partially occluded, several animals may appear in the camera frame, and the user may have only a limited amount of time for each interaction.

Main modalities: Visual graphical interface, camera-based computer vision, touch input, spoken and written questions and answers, text-to-speech output, and an immersive 3D environment for interaction with the selected animal.

3.3 Simple Task: Visual Identification

The first task allows the user to identify an animal by framing it with the device camera. Animex periodically analyzes the camera feed and, when an animal is detected, displays its species. The recognized animal becomes the active subject for the conversational and immersive tasks.

3.3.1 Hierarchical Task Analysis



3.3.2 Grounding and Error Recovery

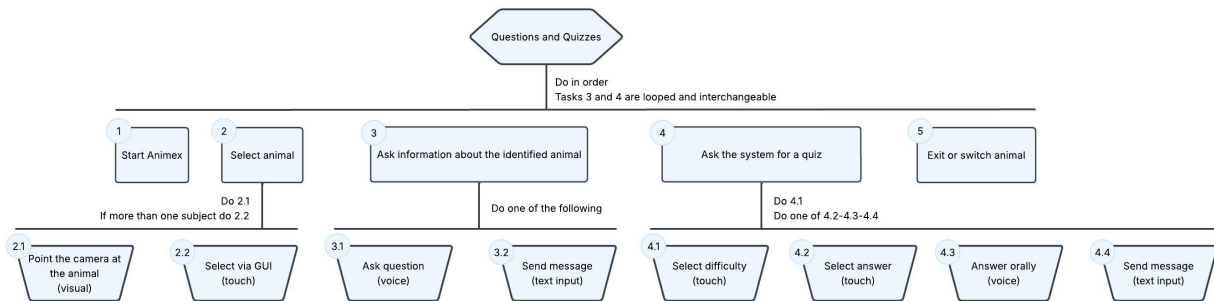
When several animals of different species are visible at the same time, Animex does not choose a referent silently. Instead, it highlights all detected animals and asks the user to select the intended subject by touch. The chosen animal is then shown as the current referent throughout the following interactions. This grounding loop prevents later answers, quizzes, or immersive content from being associated with the wrong animal species.

3.4 Core Task: Questions and Quizzes

After an animal has been identified, the user can explore it in greater depth through an interactive assistant. Questions can be asked by voice or entered as text, and answers are presented in concise, child-friendly language. The assistant maintains the selected animal as conversational context so that follow-up questions such as “*What does it eat?*” can be interpreted correctly.

The user can also request a quiz and select its difficulty. Animex generates true/false or multiple-choice questions about the current animal and provides immediate feedback after each answer. The quiz therefore reinforces information that the user has just discovered through the recognition and conversation tasks.

3.4.1 Hierarchical Task Analysis



3.4.2 Grounding and Error Recovery

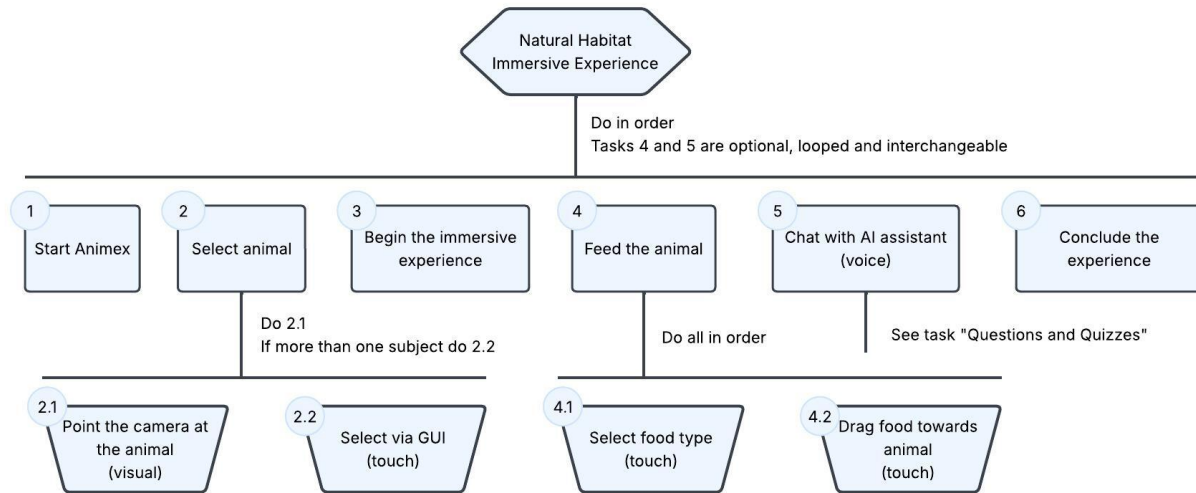
The zoo environment may make spoken questions difficult to understand. If speech recognition confidence is low, the system first asks the user to repeat the utterance and keeps the current animal visible so that the conversational referent is not lost. If the second attempt is still unclear, Animex proposes text input and preserves the partial interaction state. This repair strategy allows the user to continue without restarting the task.

3.5 Advanced Task: Natural-Habitat Immersive Experience

The advanced task places the recognized animal inside a three-dimensional representation of its natural habitat. After the user activates the immersive experience, Animex displays the animal model together with environmental elements appropriate to the species, such as vegetation and terrain. The objective is to connect factual explanations with a spatial and visual representation of the animal’s ecosystem.

The user can observe the animal from different viewpoints, interact with it by offering different types of food, and ask contextual questions through voice chat. The scene and the assistant share the same selected-animal context; therefore, the system can relate the user’s actions to explanations about diet, behavior, and habitat.

3.5.1 Hierarchical Task Analysis



3.5.2 Grounding and Error Recovery

Voice interaction in the immersive scene uses the same repair mechanism as the conversational task. When a question cannot be understood because of background noise, the system asks the user to repeat it. If recognition fails again, the user is invited to type the question. The selected animal and the active habitat remain visible during this process, maintaining a shared visual and conversational context.

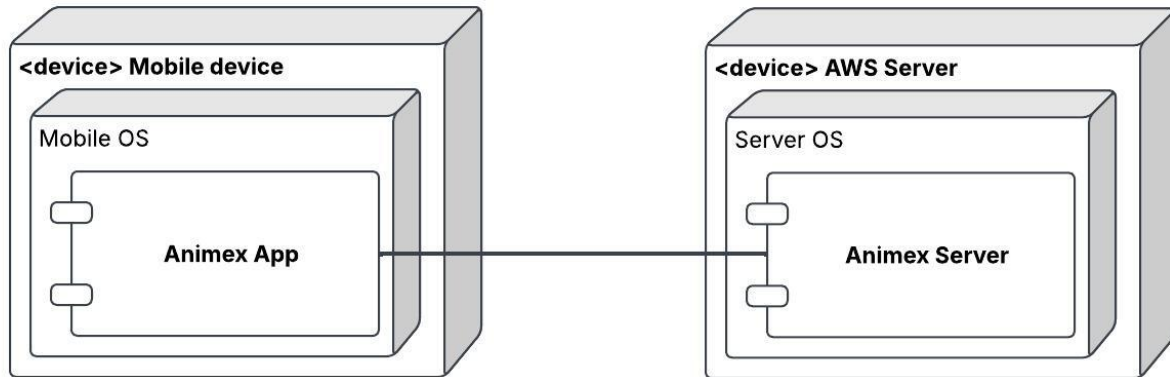
3.6 Relationship Between the Tasks

The three tasks form a progressive interaction flow. Visual identification establishes the animal that acts as the shared referent. The conversational and quiz task builds semantic knowledge about that referent, while the immersive task provides a spatial and interactive representation of the same information. The user is not forced through a rigid sequence after recognition: they may move between the assistant and the immersive experience while Animex preserves the selected animal and the interaction context.

4 Project Architecture

The Animex project is split across a two-tier architecture:

- **Front-end Mobile Application:** handles user input and displays graphical interface.
- **Back-end Server:** receives and sanitizes requests, prompts *Google Cloud*'s APIs and elaborates results for the front-end.



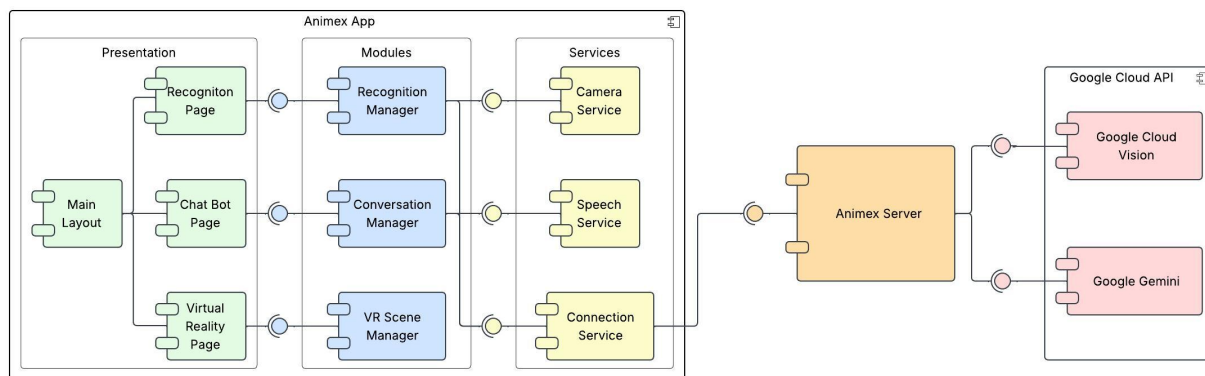
Given the aims of the MITA course, this report will focus on the architecture of the front-end application.

4.1 Front-end Application

4.1.1 Overview

The Animex application was developed utilizing the *Typescript* programming language, the *Vue*, *Ionic*, and *Three.js* graphical frameworks, the *Capacitor* input framework, and *Pinia* stores to handle global variables. The application itself can be divided into multiple layers:

- **Presentation Layer:** displays information to the user.
- **Module Layer:** handles the behavior needed for the system's tasks.
- **Service Layer:** interacts with the user's device to provide the information and functions requested by the modules.



4.1.2 Service Layer

The service layer is composed of three services:

- **Connection Service:** connects to the back-end server, sends HTTP requests and forwards the responses to the appropriate modules.
- **Camera Service:** exposes the device's camera feed.
- **Speech Service:** connects to the user's microphone to convert their speech to text and reads out system messages through text-to-speech.

4.1.3 Module Layer

The module layer is formed by three modules, one for each of the system's tasks:

- **Recognition Manager:** periodically samples the camera feed through the *Camera Service* and then sends the image to the back-end using the *Connection Service*. Once the server's response is received, it saves the recognized animal as the current conversation subject or, if multiple animals were detected, it prompts the user to select one.
- **Conversation Manager:** keeps track of the messages sent between the user and the system's LLM, utilizes the *Speech Service* to elaborate user input and system output, and provides quiz functionality.
- **VR Scene Manager:** renders the VR environment with the selected animal's model and background environment and handles the VR-entities and their behaviors.

4.1.4 Presentation Layer

The presentation layer is split across four views:

- **Main Layout:** is overlaid on top of the other views, handles navigation between pages.
- **Recognition Page:** shows the device's camera feed, highlights identified animals in frame and allows the user to ask questions and respond to quizzes via speech.
- **Chat Bot Page:** shows the message history regarding the currently selected animal, allows the user to ask questions and request quizzes both via speech and via text.
- **Virtual Reality Page:** shows the VR scene, allows users to interact with the 3D model by giving it food and to ask questions via speech.

5 Response to Heuristic Evaluation

After receiving feedback from the heuristic evaluation, improvements were made to the system. Here follows a table with the issues risen and the solutions found:

Where	Interaction Breakdown	Violated Principle	Solution	Sev.
Scanner to Assistant transition	After scanning an animal, the system does not provide a clear transition between the Scanner task and the Assistant task. While the animal is correctly recognized, users are not guided towards the next available interaction, making the different tasks appear disconnected.	H1 - Visibility of system status; H2 - Match between system and the real world; multimodal grounding of task flow	The app's flow made this issue negligible: the user scans an animal and is then free to either go and chat with the animal, or go to the VR; this choice of action is purposely left to the user, making any kind of automatic redirect unnecessary.	3
Multiple-animal recognition	The current design does not clearly specify how the system behaves when multiple animals are detected within the same camera frame. Automatically selecting one animal may lead to misunderstandings and incorrect follow-up interactions.	H5 - Error prevention; H9 - Error recovery; ambiguity handling; referent grounding	This has been fixed by giving the user a way to choose the animal when an ambiguous scenario appears.	4
Assistant interaction	During the Assistant interaction, there is no persistent visual element indicating which animal is currently being discussed. Once the recognition task is completed, users may lose track of the active referent, especially during longer conversations. This weakens multimodal grounding and may make the Assistant appear disconnected from the previous recognition task.	H1 - Visibility of system status; multimodal grounding of referents	The currently recognized animal is now always visible to the user regardless of the section of the app.	3
Assistant speech output	The assistant's spoken responses cannot currently be interrupted while they are being generated or played. Users must wait until the response finishes before continuing the interaction.	H3 - User control and freedom; interruptible grounding; turn-taking in conversational interaction	The assistant can now be correctly interrupted.	3

Where	Interaction Breakdown	Violated Principle	Solution	Sev.
Quiz access inside Assistant	The Quiz entry point is located at the top of the Assistant page. After a long conversation, users may need to scroll through a large amount of chat history before being able to start the quiz.	H6 - Recognition rather than recall; H7 - Flexibility and efficiency of use	The quiz entry point is now in a fixed position, making it more accessible to use.	2
Technical / development information exposed to users	Recognition or backend-related information useful for developers can appear meaningless or confusing to children, creating a mismatch between a playful educational app and technical system feedback.	H4 - Consistency and standards; H2 - Match between system and real world	Development related information is now correctly transformed into data that the end user can read.	2
Mobile layout and system navigation overlap	On mobile devices, the chat input bar or bottom controls may be covered by Android/iOS navigation elements or by the keyboard. This can block the main action precisely when the user wants to ask a question or answer a quiz.	H4 - Consistency and standards; H7 - Efficiency of use; accessibility principles	The UI has been redone from scratch to effectively solve this issue and improve the user's overall experience.	3
Tutorial, onboarding and contextual help	The relationship between Scanner, Assistant and AR functionalities is not immediately clear. New users may not understand that animal recognition is expected before using the other functionalities. Furthermore, the available actions inside each task are not always self-evident.	H6 - Recognition rather than recall; H10 - Help and documentation	The new UI makes this kind of information clearer: when no animal is recognized, the other actions are locked, making it clear what the user has to do. After a recognition, said actions are unlocked.	2

Where	Interaction Breakdown	Violated Principle	Solution	Sev.
Overall application architecture	Although the application includes multiple modalities such as voice interaction, image recognition and AR, these modalities currently operate as separate tasks. The Assistant relies only on voice interaction, the Scanner relies only on image recognition, and the AR experience is isolated from the other functionalities. No task combines modalities simultaneously. As a consequence, the application is multimodal at the system level but not at the interaction level.	H2 - Match between system and real world; multimodal integration principles; grounding principles	More modalities have been added to the app: for example, now all the pages allow for the use of the voice chat, making the interaction truly multimodal.	4
Quiz generation	The quiz is generated dynamically through an LLM. It is unclear whether generated questions are always restricted to the information presented to the user about the selected animal. Additionally, LLM-generated quizzes may produce repetitive question structures over time, reducing educational variety.	H5 - Error prevention; educational consistency	The AI has been improved to give less repetitive questions and to be contextualized.	3